Boundary geometry and convergence of the literal B-adaPG method

Abstract

We organize rigorous boundary results for the literal two-branch B-adaPG controller of Ou, Latafat and Themelis. An explicitly admissible Legendre kernel, flat on a boundary face, gives actual bounded nonconvergent trajectories with more than one optimal cluster point. The construction applies both to the original controller and to its prescribed fixed mild damping. Arbitrary initial stepsizes, exact asymptotic amplitudes, and the complete undamped basin classification are included. We distinguish these algorithmic counterexamples from the underlying failure of a boundary separation implication. Positive results are proved for regular composite objectives with bounded-ball strongly convex kernels, and for kernels continuous and strictly convex only on the feasible set, with an unchanged ambient controller and arbitrary closed convex constraints. We also prove cumulative-step divergence, an affine-split convergence result, a compatibility result identifying cluster gradients, and exact scalar, coupled and marginal entropy rates. All source-theorem comparisons concern arXiv:2508.01353v2; equivalence with the inaccessible journal main text is not asserted. The general unique-minimizer question outside the stated positive subclasses remains unresolved.

Keywords: Bregman proximal gradient; boundary minimizers; adaptive stepsizes; Legendre kernels; counterexamples

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1 Scope, source claims, and the exact algorithm

The reference algorithm is the original B-adaPG of [1], specifically equations (1.1), (2.3), (2.6), and Remark 2.6 of its version 2. The optimization zone is the *closed* set $K = \operatorname{cl} C$, where $C = \operatorname{int} \operatorname{dom} h$. The overbar matters: a minimizer need not be interior. The results here concern deterministic exact arithmetic, not finite-tolerance experimental stopping rules.

There are three logically distinct issues. First, same-point boundary continuity of reverse Bregman divergence need not imply separation of distinct boundary points. Second, that defect alone does not refute an algorithmic convergence theorem: an actual initialized trajectory must be constructed. Third, once such a trajectory is constructed, its exact behavior must be compared with the exact version-specific claim. We carry out these steps separately.

1.1 Notation and initialization

For $u \in \operatorname{dom} h$, $v \in C$, let

$$D_h(u, v) = h(u) - h(v) - \langle \nabla h(v), u - v \rangle, \qquad \Delta_h(u, v) = D_h(u, v) + D_h(v, u).$$

The divergence is $+\infty$ when its prescribed arguments are inadmissible. The conjugate h^* is always the *full ambient* Fenchel conjugate. For f differentiable at the relevant points, use the analogous D_f and Δ_f. Choose $x_0 \in C \cap \operatorname{dom} g$ and arbitrary $\gamma_0, \gamma_1 > 0$. The first primal step uses γ_1:

$$x_{k+1} = \operatorname*{argmin}_y \left\{ g(y) + \langle \nabla f(x_k), y - x_k \rangle + \frac{D_h(y, x_k)}{\gamma_{k+1}} \right\}. \tag{1}$$

For $k \geq 1$ with $x_k \neq x_{k-1}$ set

$$\begin{aligned} &\rho_k = \gamma_k/\gamma_{k-1}, && r_{k+1} = \sqrt{1+\rho_k}, && \delta_{k+1} = 2r_{k+1}, \\ &H_k = \nabla h - \gamma_k \nabla f, && \ell_k = \frac{\Delta_f(x_k, x_{k-1})}{\Delta_h(x_k, x_{k-1})}, && \alpha_k = \frac{D_h(x_k, x_{k-1})}{D_h(x_{k-1}, x_k)}, \end{aligned} \tag{2}$$

$$\begin{aligned} v_k &= \nabla h(x_k) + \delta_{k+1}\big(H_k(x_k) - H_k(x_{k-1})\big), \\ \Lambda_k &= \frac{2D_{h^*}(v_k, \nabla h(x_k))}{\delta_{k+1}^2 \Delta_h(x_k, x_{k-1})}. \end{aligned}$$

The undamped controller is

$$\rho_{k+1} = \min\left\{ r_{k+1}, \frac{\alpha_k}{(1+\alpha_k)2r_{k+1}[\Lambda_k - (1-\gamma_k\ell_k)]_+} \right\}, \qquad \gamma_{k+1} = \rho_{k+1}\gamma_k. \tag{3}$$

A positive numerator divided by zero means $+\infty$. The order of the conjugate arguments in (2) is essential. Both evaluations of H_k use the same γ_k.

The positive convergence results below stop when consecutive iterates agree, before evaluating a zero controller denominator, and extend the sequence constantly. This exact stopping convention is explicitly part of those statements. The counterexample trajectories never repeat, so no completion of a zero-denominator branch is needed for them.

For fixed damping $\kappa \in [99/100, 1)$, the source modification in Remark 2.8 multiplies the *entire minimum* in (3) by κ. It does not change r_{k+1} or δ_{k+1}. In particular δ_{k+1} is not $2\rho_{k+1}$ in the damped rule.

1.2 Source hypotheses and limits of the comparison

The kernel hypotheses used throughout are: h is proper, lower semicontinuous, convex, 1-coercive and Legendre; $C \neq \varnothing$; $h \in C^2(C)$ with positive definite Hessian. Here 1-coercive means $h(x)/\|x\| \to +\infty$ as $\|x\| \to \infty$, including extended-real values. The additional boundary assumptions are

$$D_h(z, y_i) \longrightarrow 0 \quad \text{whenever } z \in K,\ C \ni y_i \longrightarrow z, \tag{4}$$

$$y \longmapsto D_h(z, y) \text{ is level bounded for every } z \in K \setminus C. \tag{5}$$

These are Assumption 2.4 of [1]; they do not assert that vanishing divergence separates arbitrary boundary points. Local relative smoothness means that for each compact convex $E \subset C$ some $L_E > 0$ makes $L_E h - f$ convex on E.

Under its printed assumptions, Theorem 2.7(ii) of [1] asserts boundedness and a *unique optimal limit point*. Nonconvergence alone would not contradict that wording; two distinct optimal cluster points do. Equation (2.8), the best-objective-value assertion, is not contradicted by our examples. Theorem 2.7(i), which assumes an interior solution, is not tested by them. Remark 2.8's undamped full-convergence assertion is a conjecture, whereas its fixed-damping full-convergence assertion is accompanied by a proof in Appendix A.

A journal article with matching bibliographic identity exists [2], but its main theorem and hypotheses were not accessible in the inspected subscription preview. The institutional accepted-manuscript record [3] was explicitly embargoed until 15 September 2027; that embargo was respected. Accordingly no statement here claims a counterexample to an uninspected journal theorem. The source comparison is exclusively with the inspected arXiv version 2, revised 5 June 2026.

2 An admissible flat-face kernel

Let

$$\varepsilon = \frac{1}{256}, \qquad a = \frac{1}{100}, \qquad K = [-1,1] \times [0,\varepsilon], \qquad C = \operatorname{int} K, \tag{6}$$

and define

$$b(x) = (1+x)\log(1+x) + (1-x)\log(1-x), \quad q(y) = -\sqrt{y} - \sqrt{\varepsilon - y},$$

with $0 \log 0 = 0$. Our principal kernel is

$$h(x,y) = \begin{cases} y\{b(x) - ax\sin(\log y)\} + q(y), & (x,y) \in K,\ y > 0, \\ -\sqrt{\varepsilon}, & -1 \le x \le 1,\ y = 0, \\ +\infty, & (x,y) \notin K. \end{cases} \tag{7}$$

It is constant on the entire bottom edge, despite being strictly convex in the interior.

Lemma 1. *The kernel* (7) *is proper, closed, convex, 1-coercive and Legendre, continuous on K, and C^∞ with positive definite Hessian on C. It satisfies* (4)–(5) *at every point of K. Moreover $\nabla h : C \to \mathbb{R}^2$ is a smooth bijection with smooth inverse, and h^* is finite and smooth everywhere.*

Proof. Continuity follows from boundedness of b and $y \sin \log y \to 0$. The closed-domain extension is lower semicontinuous and proper; bounded effective domain implies 1-coercivity. Write $t = \log y$. Direct differentiation gives

$$\begin{aligned} h_x &= y[b'(x) - a\sin t], & h_y &= b(x) - ax(\sin t + \cos t) + q'(y), \\ h_{xx} &= yb''(x), & h_{xy} &= b'(x) - a(\sin t + \cos t), \\ h_{yy} &= q''(y) - \frac{ax}{y}(\cos t - \sin t), & q''(y) &= \tfrac{1}{4}[y^{-3/2} + (\varepsilon - y)^{-3/2}]. \end{aligned}$$

Put $u = \operatorname{arctanh} x$, $B = b''(x) = 2\cosh^2 u$, and $\beta = b'(x) = 2u$. Since $\cosh u \geq 1 + u^2/2 \geq \sqrt{2}|u|$, one has $\beta^2/B \leq 1$. With $c = a(\sin t + \cos t)$,

$$\frac{(\beta - c)^2}{B} \leq 2 + 2a^2, \qquad yh_{yy} \geq \frac{1}{4\sqrt{y}} - 2a \geq 4 - 2a.$$

Thus $yh_{yy} - h_{xy}^2/B \geq 2 - 2a - 2a^2 > 0$. Together with $h_{xx} > 0$, this proves positive definiteness. Convexity extends to K by contracting any boundary endpoints toward one fixed interior point and passing to the limit. Interior strict convexity follows as well.

As $y \downarrow 0$, $h_y \to -\infty$; as $y \uparrow \varepsilon$, $h_y \to +\infty$, uniformly in the other bounded terms. These facts cover mixed-corner approaches. On either vertical side with limiting height in $(0, \varepsilon)$, h_x diverges in magnitude. Thus essential smoothness holds. At a horizontal edge an inward square-root difference quotient tends to $-\infty$; at an open vertical edge the entropy's inward difference quotient does likewise. Hence ∂h is empty on the boundary and $\operatorname{dom} \partial h = C$. Strict convexity on C proves essential strict convexity.

For $Z = (X, Y) \in K$ and $(u, v) \in C$ tending to Z, it suffices by continuity to check

$$h_x(u, v)(X - u) \to 0, \qquad h_y(u, v)(Y - v) \to 0.$$

For the first product use bounded v, bounded oscillatory coefficient, and $b'(u)(X - u) \to 0$. At $X = \pm 1$ the last assertion is the elementary $s \log s \to 0$ limit. For the second product only $q'(v)$ can be unbounded. Its product with $Y - v$ is $O(\sqrt{v}) + O(v)$ at $Y = 0$, and $O(\sqrt{\varepsilon - v}) + O(\varepsilon - v)$ at $Y = \varepsilon$; otherwise it vanishes by continuity. This proves (4) for arbitrary approaches. All reverse-divergence sublevels lie in bounded C, proving (5).

Finally, $h(z) - \langle p, z \rangle$ attains its minimum on compact K for every finite p. The inward difference quotients exclude boundary minima. Interior strict convexity gives uniqueness and $\nabla h(z) = p$. The positive Hessian and inverse function theorem give the asserted smooth inverse and conjugate properties. ∎

2.1 The structural separation error

The simpler kernel $h_0(x, y) = yb(x) + q(y)$ on K, with the same bottom extension, is obtained by putting $a = 0$ in the preceding proof. Thus it has all the same source properties. For every bottom point $Z = (X, 0)$ and $w_v = (0, v)$,

$$D_{h_0}(Z, w_v) = -\sqrt{\varepsilon} - q(v) + vq'(v) = \tfrac{1}{2}\sqrt{v} + O(v) \longrightarrow 0. \tag{8}$$

If $X \neq 0$, $w_v \to (0, 0) \neq Z$. Thus the asserted boundary convergence-transfer implication in Fact 1.2(iii) of [1], when read at this generality, is false. This is an auxiliary-fact counterexample, not by itself an optimization trajectory. In particular, a proof gap must not be confused with a theorem counterexample. The next sections supply the actual trajectory.

3 Actual undamped trajectories and arbitrary step-sizes

Take $f(x,y) = y$ and $g = 0$ on all $\mathbb{R}^2$. Every positive L makes $Lh - f$ convex, so local relative smoothness holds. The solution set over K is $[-1,1] \times \{0\}$. Write

$$\Phi = \frac{1+\sqrt{5}}{2}, \qquad X(y) = \tanh\Big(\frac{a}{2}\sin\log y\Big).$$

Theorem 1. *Initialize the literal undamped method with any $0 < y_0 \le \varepsilon/8$, $x_0 = X(y_0)$, and any $\gamma_0, \gamma_1 > 0$. Then every iterate $z_k = (x_k, y_k)$ is defined and interior, y_k strictly decreases, and the second controller branch is exactly $+\infty$ at every adaptive index. In particular*

$$\rho_{k+1} = \sqrt{1+\rho_k}.$$

Let $t_1 = \gamma_1/\gamma_0$, $t_{j+1} = \sqrt{1+t_j}$ and define

$$P = \prod_{j=2}^{\infty} \frac{t_j}{\Phi}, \qquad C_\gamma = \gamma_1 P, \qquad T = \Phi C_\gamma, \qquad A = \frac{1}{4T^2}, \qquad \beta = \log A.$$

The product is finite and strictly positive. For a finite constant R_0,

$$\begin{aligned} \rho_k &= \Phi + O(\Phi^{-2k}), & \gamma_k &= C_\gamma \Phi^{k-1}(1 + O(\Phi^{-2k})), \\ \partial_y h(z_k) &= -T\Phi^k + R_0 + O(\Phi^{-k}), & & \\ y_k &= A\Phi^{-2k}(1 + O(\Phi^{-k})), & \log y_k &= \beta - 2k\log\Phi + O(\Phi^{-k}), \\ x_k &= \tanh\Big[\frac{a}{2}\sin\big(\beta - 2k\log\Phi + O(\Phi^{-k})\big)\Big]. & & \end{aligned} \tag{9}$$

The bounded trajectory does not converge. Its full cluster set is $[-\tanh(a/2), \tanh(a/2)] \times \{0\}$, and all cluster points are minimizers. Objective values converge to zero.

Proof. The entropy conjugate is $J(p) = 2\log\cosh(p/2)$. Minimizing $h(x,y)$ in x gives the reduced kernel

$$H(y) = q(y) - yJ(a\sin\log y), \qquad H'(y) = h_y(X(y), y).$$

Its second derivative is the positive Hessian Schur complement, and H' increases from $-\infty$ to $+\infty$. Let $Y = (H')^{-1}$ and $F = H^*$. Then

$$h^*(0,r) = F(r), \quad \nabla h^*(0,r) = (X(Y(r)), Y(r)), \quad F''(r) = 1/H''(Y(r)) > 0.$$

The actual full primal step is $\nabla h(z_k) = \nabla h(z_{k-1}) - \gamma_k(0,1)$. Starting with horizontal dual coordinate zero, it remains zero. With $r_k = H'(y_k)$,

$$r_k = r_{k-1} - \gamma_k, \qquad y_k = Y(r_k), \qquad x_k = X(y_k). \tag{10}$$

Every finite positive step decreases y and stays in the initial strip.

Put $L(t) = 2\log\cosh((a/2)\sin t)$. Differentiation gives

$$H'''(y) = q'''(y) + \frac{L'(\log y) - L'''(\log y)}{y^2}.$$

Writing $A_0 = a/2$, elementary derivative bounds yield

$$|L'| \le 2A_0^2, \qquad |L'''| \le 4A_0^4 + 8A_0^2, \qquad |L' - L'''| < 1/1000.$$

Indeed, for $u = A_0 \sin t$,

$$L''' = 2[-2\operatorname{sech}^2 u \tanh u\,(u')^3 + 3\operatorname{sech}^2 u\, u'u'' + \tanh u\, u'''].$$

On $0 < y \le \varepsilon/8$, $y/(\varepsilon - y) \le 1/7$ and

$$q'''(y) = \tfrac{3}{8}[-y^{-5/2} + (\varepsilon - y)^{-5/2}] < -\tfrac{3}{16}y^{-5/2}.$$

Consequently $H''' < 0$ there, so F'' is strictly increasing on the entire dual ray below r_0. This includes every extrapolated controller point.

At adaptive index k, write $r = r_k$, $\gamma = \gamma_k$, $\delta = \delta_{k+1}$. Constancy of ∇f gives $\ell_k = 0$ and

$$\Delta_h(z_k, z_{k-1}) = \gamma[Y(r + \gamma) - Y(r)] > \gamma^2 F''(r) > 0.$$

Both directed divergences are strictly positive. The exact ordered conjugate probe is

$$\begin{aligned} D_{h^*}(v_k, \nabla h(z_k)) &= D_F(r - \delta\gamma, r) \\ &= \int_0^{\delta\gamma} (\delta\gamma - s)F''(r - s)\, ds < \tfrac{1}{2}\delta^2\gamma^2 F''(r). \end{aligned}$$

Thus $0 < \Lambda_k < 1$ for every positive δ. The second branch is exactly infinite, proving the stated growth recursion and well-definedness inductively. No geometric growth was assumed to establish its own branch selection.

The exact error identity

$$t_{j+1} - \Phi = \frac{t_j - \Phi}{t_{j+1} + \Phi}$$

and $t_{j+1} > 1$ imply $|t_j - \Phi| \le |t_1 - \Phi|\Phi^{-2(j-1)}$. The product defining P therefore converges positively; its logarithmic tail is $O(\Phi^{-2k})$. It follows that the stepsize estimate in (9) holds. The errors $e_j = \gamma_j - C_\gamma \Phi^{j-1} = O(\Phi^{-j})$ are summable. Summing (10) gives

$$r_k = -T\Phi^k + \left(r_0 + T - \sum_{j \ge 1} e_j\right) + O(\Phi^{-k}).$$

On the strip,

$$H'(y) = -\frac{1}{2\sqrt{y}} + \frac{1}{2\sqrt{\varepsilon - y}} - L(\log y) - L'(\log y) = -\frac{1}{2\sqrt{y}} + O(1).$$

Inverting gives all remaining asymptotics, including the exact amplitude and phase. The leading constant depends on the initial steps, but not on y_0.

For elementary nonconvergence put $\omega = 2\log\Phi \in (0,\pi)$ and $s_k = \sin(\beta - k\omega)$. If s_k converged, its recurrence $s_{k+2} - 2\cos\omega\, s_{k+1} + s_k = 0$ would force the limit to be zero. The sine addition formula would then force $\cos(\beta - k\omega) \to 0$, a contradiction. A vanishing phase perturbation does not alter convergence, and tanh has a continuous inverse on the relevant compact interval. Hence x_k cannot converge.

For the full cluster interval we use the classical Gelfond–Schneider theorem, including its complex-exponent form [4, 5]. If $\log\Phi/\pi = q \in \mathbb{Q}$, then $q \neq 0$ and the branch $\log(-1) = i\pi$ gives $(-1)^{-iq} = \exp(q\pi) = \Phi$, contrary to the theorem because $-iq$ is algebraic and not rational. Thus $\log\Phi/\pi$ is irrational. Irrational-rotation density gives every circle angle as a subsequential limit of $\beta - k\omega$; vanishing phase errors preserve this. The continuous image under $\theta \mapsto \tanh((a/2)\sin\theta)$ is precisely the asserted interval. Since $y_k \to 0$, it is exactly the full cluster set. Every such point is optimal. ∎

Corollary 3.1. *For $y_0 = \varepsilon/8$, $x_0 = X(y_0)$, $\gamma_1 = 1/(2\Phi)$ and $\gamma_0 = 1/(2\Phi^2)$, the exact identities are*

$$\rho_k = \Phi, \quad \gamma_k = \frac{\Phi^{k-1}}{2\Phi}, \quad r_k = r_0 - \frac{\Phi^k - 1}{2}.$$

In (9), $A = 1$ and $\beta = 0$.

The same amplitude and phase hold for any allowed balanced y_0 with these two step-sizes. Scaling both initial steps by $c > 0$ multiplies A by c^{-2} and shifts β by $-2\log c$.

4 The complete undamped basin classification

Theorem 2. *For the same kernel and objective, start the undamped algorithm from any $z_0 = (x_0, y_0) \in C$ and any positive γ_0, γ_1. Define*

$$p = h_x(z_0) = y_0 \left[\log\frac{1+x_0}{1-x_0} - a\sin\log y_0\right].$$

The trajectory is well defined at every finite index,

$$h_x(z_k) = p, \qquad x_k = \tanh\left(\frac{p/y_k + a\sin\log y_k}{2}\right), \qquad h_y(z_k) = h_y(z_0) - \sum_{j=1}^{k}\gamma_j,$$

and $\sum_j \gamma_j = \infty$, $y_k \downarrow 0$. If $p > 0$ it converges to $(1,0)$; if $p < 0$ it converges to $(-1,0)$. If $p = 0$ it is bounded and nonconvergent. Hence the exceptional nonconvergent basin is exactly the graph $x = X(y)$, $0 < y < \varepsilon$. The two corner basins are open, and their union is dense and has full planar Lebesgue measure in C. In all cases objective values converge to zero and all cluster points are optimal.

Proof. For fixed p solve $h_x = p$ to obtain $X_p(y) = \tanh((p/y + a\sin\log y)/2)$. Define

$$H_p(y) = \min_x[h(x,y) - px] = q(y) - yJ(p/y + a\sin\log y).$$

Envelope differentiation gives $H_p' = h_y(X_p(y), y)$ and $H_p'' = h_{yy} - h_{xy}^2/h_{xx} > 0$. Because b and the other terms in $h_y - q'$ are uniformly bounded, H_p' ranges from $-\infty$ to $+\infty$. Let $Y_p = (H_p')^{-1}$ and $F_p(r) = h^*(p,r)$. The full inverse gradient is $(X_p(Y_p(r)), Y_p(r))$, and $F_p'' > 0$.

The primal update preserves p and subtracts γ_k from $r_k = h_y(z_k)$. On this curve the Bregman divergences reduce exactly to those of H_p; equivalently their dual orientations are reversed:

$$D_h(z_k, z_{k-1}) = D_{F_p}(r_{k-1}, r_k), \qquad D_h(z_{k-1}, z_k) = D_{F_p}(r_k, r_{k-1}).$$

Both are positive and finite. The controller probe is exactly $D_{F_p}(r_k - \delta_{k+1}\gamma_k, r_k)$, so all controller quantities are finite with positive selected ratio. This proves well-definedness without assuming any branch is active.

Let $R = \max\{\rho_1, \Phi\}$. The first branch implies $0 < \rho_k \le R$, $1 < r_{k+1} \le R$, and $2 < \delta_{k+1} \le 2R$. If the step sum were finite, $r_k \to r_\infty \in \mathbb{R}$ and $\gamma_k \to 0$. Every previous-step and extrapolated dual interval would shrink to r_∞. Uniform Taylor expansion using $c = F_p''(r_\infty) > 0$ gives

$$\begin{aligned} D_{F_p}(r_{k-1}, r_k) &= \tfrac{1}{2}c\gamma_k^2(1 + o(1)), \\ D_{F_p}(r_k, r_{k-1}) &= \tfrac{1}{2}c\gamma_k^2(1 + o(1)), \\ 2D_{F_p}(r_k - \delta_{k+1}\gamma_k, r_k) &= c\delta_{k+1}^2\gamma_k^2(1 + o(1)). \end{aligned}$$

The last error is uniform because δ_{k+1} is bounded. Thus $\alpha_k \to 1$ and $\Lambda_k \to 1$, while $\ell_k = 0$. The second branch tends to $+\infty$, so eventually the first branch is selected and $\gamma_{k+1} > \gamma_k > 0$. This contradicts $\gamma_k \to 0$. Therefore $r_k \to -\infty$ and $y_k \downarrow 0$.

For $p \neq 0$ the explicit X_p gives the asserted corner limit; indeed

$$|x_k - \operatorname{sign} p| \le 2\exp(a - |p|/y_k).$$

For $p = 0$, some finite N has $y_N \le \varepsilon/8$. Restarting at this point with initial steps γ_N, γ_{N+1} reproduces the actual tail exactly, including its first outgoing step and first adaptive index. Theorem 1 gives nonconvergence. Finally, strict monotonicity of b' gives $\operatorname{sign} p = \operatorname{sign}(x_0 - X(y_0))$. The graph is continuous, has empty interior, and has measure zero by its singleton horizontal sections and Fubini's theorem. This proves the basin assertions. ∎

5 Fixed mild damping and the precise theorem conflicts

Theorem 3. *Fix $\kappa \in [99/100, 1)$, let*

$$R = \frac{\kappa^2 + \sqrt{\kappa^4 + 4\kappa^2}}{2}, \qquad y_0 = \varepsilon/8, \qquad x_0 = X(y_0),$$

and use the literal damped controller with any $\gamma_0, \gamma_1 > 0$. Every iterate is defined and interior; the second branch is infinite and $\rho_{k+1} = \kappa\sqrt{1+\rho_k}$. Put $q = \kappa/2$ and

$$C_\gamma = \gamma_1 \prod_{j=2}^{\infty} \frac{\rho_j}{R}, \qquad B = \left(\frac{R-1}{2C_\gamma}\right)^2.$$

Then $C_\gamma \in (0, \infty)$ and

$$\rho_k = R + O(q^{k-1}), \qquad \gamma_k = C_\gamma R^{k-1}(1 + O(q^k)),$$
$$r_k = -\frac{C_\gamma}{R-1} R^k + O(1), \qquad y_k = B R^{-2k}(1 + O(R^{-k})),$$
$$x_k = \tanh\Big[\frac{a}{2} \sin\big(\log B - 2k \log R + O(R^{-k})\big)\Big].$$

The bounded primal trajectory is nonconvergent and has at least two distinct optimal cluster points. Nevertheless

$$\gamma_k y_k \to 0, \qquad \gamma_k y_{k-1} \to 0, \qquad D_h(z_k, z_{k-1}) \to 0,$$

and $D_h(Z, z_k) \to 0$ for every fixed bottom point $Z = (X, 0)$.

Proof. The invariant strip and strict $\Lambda_k < 1$ argument in Theorem 1 hold for every positive extrapolation length and require no fixed-ratio hypothesis. Multiplying the minimum by κ therefore gives exactly the claimed ratio recursion. The positive fixed point solves $R^2 = \kappa^2(1 + R)$. Since $\kappa^2 > 1/2$ and $\kappa < 1$, $1 < R < \Phi < 2$.

For $T(t) = \kappa\sqrt{1+t}$, $0 < T'(t) \le q < 1/2$ on $[0, \infty)$. Thus $|\rho_k - R| \le q^{k-1}|\rho_1 - R|$. The positive product converges, and its logarithmic tail gives the stepsize expansion. Since $Rq < 1$, the series of differences $\gamma_j - C_\gamma R^{j-1}$ is absolutely summable. Summing the exact dual update and using $H'(y) = -1/(2\sqrt{y}) + O(1)$ proves all asymptotic formulas. The sine-recurrence argument of Theorem 1 applies to the arbitrary phase $\log B$ and angle $2\log R \in (0, \pi)$, proving nonconvergence without an irrationality assumption. No full cluster interval is claimed for this damped family.

The displayed scalar vanishings follow from $\gamma_k = O(R^k)$, $y_k = O(R^{-2k})$, and

$$0 < D_h(z_k, z_{k-1}) < \Delta_h(z_k, z_{k-1}) = \gamma_k(y_{k-1} - y_k) = O(R^{-k}).$$

For any $Z = (X, 0)$ the horizontal dual component is zero, so

$$\begin{aligned} D_h(Z, z_k) &= -\sqrt{\varepsilon} - H(y_k) + y_k H'(y_k) \\ &= -\sqrt{\varepsilon} - q(y_k) + y_k q'(y_k) - y_k L'(\log y_k) \\ &= \tfrac{1}{2}\sqrt{y_k} + O(y_k) \longrightarrow 0. \end{aligned}$$

This expression is independent of X. ∎

For the damping-dependent special initialization $\gamma_1 = (R-1)/2$, $\gamma_0 = (R-1)/(2R)$, one has exactly

$$\rho_k = R, \quad \gamma_k = (R-1)R^{k-1}/2, \quad \delta_{k+1} = 2R/\kappa, \quad r_k = r_0 - (R^k - 1)/2,$$

and $B = 1$. Alternatively, the *same* two stepsizes in Corollary 3.1 may be used for every allowed damping factor, by Theorem 3.

5.1 What is disproved, and what remains valid

All source assumptions have been checked for the actual trajectories. The undamped trajectory has multiple distinct *optimal* cluster points, contradicting the unique-optimal-limit-point conclusion of Theorem 2.7(ii) in [1]. It also disproves that version's undamped full-convergence conjecture in Remark 2.8. It preserves boundedness and objective convergence.

The damped construction separately contradicts the primal convergence assertion for fixed multiplicative damping in Remark 2.8 and Appendix A of the same version. Its damping lies inside the stated small-damping range, since $0 < 1-\kappa \leq 1/100 < (2-\sqrt{2})/2$. It does not contradict the scalar decay claims, which hold explicitly. The final implication from $D_h(Z, z_k) \to 0$ to primal convergence fails because the kernel does not separate the bottom boundary points. This is an actual theorem counterexample together with an identified proof failure; neither statement is inferred merely from the other. No conclusion is drawn about another damping schedule, B-adaPG$_\alpha$, an interior-minimizer theorem, or an uninspected journal statement.

6 An energy inequality for regular composites

We now state a common regular setting. Let h satisfy the kernel hypotheses and (4)–(5), and assume $K = \operatorname{dom} h = \operatorname{cl} C$. Let $U \supset K$ be open and convex, $f : U \to \mathbb{R}$ convex and C^1, locally relatively smooth on compact convex subsets of C. Let $g_0 : \mathbb{R}^n \to \mathbb{R}$ be finite convex, let D be nonempty closed convex with $D \cap C \neq \varnothing$, and set

$$S = K \cap D, \qquad g = g_0 + I_D, \qquad F = f + g_0 \text{ on } S.$$

Assume F attains its minimum m on S. Initialize at any $x_0 \in C \cap D$ with arbitrary positive stepsizes and use (1)–(3), with the stated stopping convention. In particular the primal minimization is over S, but the conjugate and gradients in the controller remain ambient.

For exact source applicability, replace f outside U by

$$f_{\text{ext}}(y) = \sup_{a \in U} \{f(a) + \langle \nabla f(a), y - a \rangle\}. \tag{11}$$

This is a proper closed convex function agreeing with f on U, by the support inequality and equality at $a = y$. It changes no feasible value or algorithm quantity.

Lemma 2. *In the regular setting, every primal update is uniquely defined in $C \cap D$, and every nonstopped stepsize is finite positive. For $k \geq 2$, every minimizer $a \in S$, and $P_k = F(x_k) - m$, put*

$$d_k = 1 - \frac{\rho_k}{2r_k}, \qquad V_k(a) = D_h(a, x_k) + \gamma_k(1 + r_k)P_{k-1} + d_k D_h(x_k, x_{k-1}).$$

Then $d_k \geq 1/2$ and

$$\begin{aligned} V_{k+1}(a) &\leq V_k(a) - A_k P_{k-1} - B_k D_h(x_k, x_{k-1}) - \gamma_{k+1} D_f(a, x_k), \\ A_k &= \gamma_k[1 + r_k - \rho_{k+1} r_{k+1}] \geq 0, \\ B_k &= d_k - \rho_{k+1} r_{k+1} \frac{1 + \alpha_k}{\alpha_k}[\Lambda_k - (1 - \gamma_k \ell_k)] \geq 0. \end{aligned} \tag{12}$$

The bracket in B_k is signed. The orbit is bounded, has an optimal cluster point, and

$$\sum_{k \geq 2} \gamma_{k+1} D_f(a, x_k) < \infty. \tag{13}$$

No boundary separation or uniqueness of optimal clusters is asserted in this lemma.

Proof. For a current $x \in C$ and step $\gamma > 0$, the scaled proximal objective is

$$h(y) + \gamma g_0(y) + I_D(y) + \langle \gamma \nabla f(x) - \nabla h(x), y \rangle.$$

An affine support of g_0 and 1-coercivity of h make it coercive. It is proper and lower semicontinuous, so a minimizer exists. The sum rule applies because $D \cap C \neq \varnothing$; equivalently $\operatorname{ri} D \cap C \neq \varnothing$, by approximation within the open set C. At a minimizer the subdifferential of h is nonempty, so essential smoothness forces interiority. Interior strict convexity gives uniqueness. There are actual selections $\xi_{k+1} \in \partial g_0(x_{k+1})$ and $n_{k+1} \in N_D(x_{k+1})$ such that, writing $p_k = \nabla h(x_k)$,

$$p_{k+1} - p_k = -\gamma_{k+1}[\nabla f(x_k) + \xi_{k+1} + n_{k+1}]. \tag{14}$$

Normals need not be bounded. Full-domain Legendre duality gives $\operatorname{dom} h^* = \mathbb{R}^n$ and continuous ∇h^*. Thus every ambient dual probe is valid, and strict interior convexity makes all nonzero-step denominators positive. The first branch is finite positive, the second positive or infinite. A repeated iterate in (14) is optimal. Conversely an S-optimal interior iterate is locally D-optimal, hence convex first-order optimality makes the next proximal point identical.

For completeness, the source ingredient retained here is Lemma 3.1 of [1], its exact three-point/subgradient identity, before the nonnegative smooth remainder is discarded.

Its cross term is bounded by the ambient Bregman–Young inequality, equation (3.11) there:

$$B_{\mathrm{cross},k+1} \le \frac{\rho_{k+1}}{\delta_{k+1}} D_h(x_{k+1}, x_k) + \frac{\delta_{k+1}\rho_{k+1}}{2} \Lambda_k \Delta_h(x_k, x_{k-1}).$$

Retain $-\gamma_{k+1} D_f(a, x_k)$, discard only the other nonnegative convex subgradient slacks, and substitute $\vartheta_{k+1} = r_{k+1}$ and $\delta_{k+1} = 2r_{k+1}$. Collecting the divergence and objective coefficients gives exactly (12). This derivation works for extended-valued $g_0 + I_D$: all selected-subgradient inequalities are between feasible points. The first branch gives $\rho_k \le r_k$ for $k \ge 2$ and $\rho_{k+1} r_{k+1} \le 1 + \rho_k \le 1 + r_k$. For a positive signed bracket the second branch bounds the term subtracted from d_k by $1/2$; for a nonpositive bracket it only increases B_k. Hence all stated coefficients are nonnegative. Beginning at $k = 2$ imposes no false bound on arbitrary ρ_1.

For a minimizer, $V_k(a)$ is nonnegative and nonincreasing. It bounds $D_h(a, x_k)$ and proves (13). Reverse level boundedness gives boundedness of the orbit. For an interior comparison point this is the ordinary interior Legendre property; for a boundary minimizer it is (5).

We also record an independent justification of the optimal-subsequence step, to avoid importing the disputed boundary uniqueness assertion. Suppose $\inf_k F(x_k) > m$. For a minimizer a choose $q = (1-t)a + t x_0 \in C \cap D$ with small $t > 0$, so that $F(q) < \inf_k F(x_k)$. The same merit identity applies with $P_k(q) = F(x_k) - F(q) \ge \eta > 0$. It bounds γ_k and $D_h(q, x_k)$. The latter traps all iterates in a compact subset of C: if $p = \nabla h(x)$, a small ball about q inside C yields

$$D_h(q, x) = h(q) + h^*(p) - \langle q, p \rangle \ge r\|p\| + h(q) - M,$$

where M bounds h on that ball. Thus p is bounded, and the continuous inverse ∇h^* maps the resulting compact dual ball into a compact subset of C.

If $\gamma_k \to 0$, on a slightly larger compact convex interior set the Hessian of h has positive lower and finite upper bounds, ∇f is Lipschitz by local relative smoothness, and ∂g_0 is uniformly bounded. Pairing (14) with $x_k - x_{k-1}$ uses the favorable normal sign to give

$$\Delta_h(x_k, x_{k-1}) \le \gamma_k M_1 \|x_k - x_{k-1}\|,$$

so these differences tend to zero. Uniform Taylor expansion, with $A = \nabla^2 h(x_k)$, $d = x_k - x_{k-1}$ and $u = \nabla h(x_k) - \nabla h(x_{k-1})$, gives

$$u = Ad + o(\|d\|), \quad \Delta_h = d^T A d + o(\|d\|^2), \quad \alpha_k \to 1.$$

The ratio bound $R = \max\{\rho_1, \Phi\}$ gives $2 \le \delta_{k+1} \le 2R$. Since $\nabla^2 h^*(p_k) = A^{-1}$, the ordered ambient Taylor expansion gives $\Lambda_k \to 1$ and $\gamma_k \ell_k \to 0$. Hence the second branch tends to infinity, so eventually $\gamma_{k+1} > \gamma_k$, a contradiction. Therefore $\sum \gamma_k = \infty$ in this

hypothetical setting. The boundedness of γ_k, r_k and telescoping identity

$$\sum_{k=2}^{N} A_k = \sum_{k=2}^{N} \gamma_k + \gamma_2 r_2 - \gamma_{N+1} r_{N+1} \longrightarrow \infty$$

contradict summation of (12) with gaps at least η. Thus $\inf_k F(x_k) = m$. A nonstopped run has no optimal iterate, so the minimizing subsequence can be chosen at indices tending to infinity. Boundedness, continuity of F on S, and closedness of S supply an optimal cluster point. No form of Fact 1.2(iii) is used. ∎

7 Positive convergence from feasible-set geometry

Theorem 4. *In the regular setting of the previous section, additionally assume that $h|_S$ is continuous in the relative topology of S and strictly convex on S. Then the full original ambient undamped B-adaPG sequence converges in Euclidean norm to a minimizer of F on S.*

No strict convexity of the objective, unique minimizer, boundary avoidance, bounded constraint normals, bounded-ball Hessian lower bound, or stepsize floor is required. No rate is asserted. The added kernel geometry is required only on S; the original ambient kernel and boundary assumptions remain in force on all of K.

Proof. The stopped case follows from Lemma 2. Otherwise let Q be S intersected with a large closed ball containing the bounded orbit, its clusters, and the comparison points in use. Define the Jensen gap

$$J(u,v) = \frac{h(u)+h(v)}{2} - h\left(\frac{u+v}{2}\right), \qquad u, v \in Q.$$

It is continuous and positive off the diagonal. For each $\eta > 0$ with nonempty separated-pair set,

$$j(\eta) = \min_{u,v \in Q,\, \|u-v\| \geq \eta} J(u,v) > 0, \qquad D_h(u,v) \geq 2J(u,v) \quad (v \in C \cap Q). \tag{15}$$

The last inequality is the support inequality at v applied to $(u+v)/2$. No boundary gradient is used.

If a, b are optimal clusters, $V_k(a) - V_k(b) = D_h(a, x_k) - D_h(b, x_k)$ has a limit. Along a subsequence tending to a, (4) makes that limit nonpositive; along one tending to b it is nonnegative. It is therefore zero, and $D_h(b, x_k) \to 0$ along the first subsequence. Equation (15) forces $a = b$. Denote the unique *optimal* cluster by z; we have not yet shown that all clusters are optimal.

For any feasible fixed anchor a, the three-point identity and (14) give

$$\begin{aligned} D_h(a, x_{k+1}) - D_h(a, x_k) \leq & -D_h(x_{k+1}, x_k) \\ & + \gamma_{k+1}\{\langle \nabla f(x_k), a - x_{k+1} \rangle + g_0(a) - g_0(x_{k+1})\}. \end{aligned} \tag{16}$$

The normal term is nonpositive by feasibility. On Q, bound ∇f by B and let $T = 1 + B \operatorname{diam} Q + \max_Q g_0 - \min_Q g_0 > 0$. Hold $a = x_p$ fixed and telescope to obtain

$$D_h(x_p, x_q) \le T \sum_{i=p}^{q-1} \gamma_{i+1}. \tag{17}$$

Thus a displacement at least η costs at least $2j(\eta)/T$ in cumulative steps. The exiting step is included.

If $\sum \gamma_k < \infty$, (17) and (15) make the orbit Cauchy, hence convergent to its optimal cluster z. Suppose instead the sum is infinite. Put $a(x) = D_f(z, x) \ge 0$. It is continuous on Q, and (13) holds. If a cluster y had $a(y) > 0$, choose a ball of radius r about y on which $a \ge \eta > 0$. At arbitrarily late visits within radius $r/2$, any first exit from the r-ball costs at least $2j(r/2)/T$ in steps, all charged at old endpoints where $a \ge \eta$. This contradicts vanishing tails of (13). If there is no exit, the infinite remaining step sum gives the same contradiction. Hence $a(y) = 0$ at every cluster.

The affine support of f at such a y touches f at z. Since f is differentiable on an open neighborhood of z, that support has slope $c = \nabla f(z)$. Thus

$$\nabla f(y) = c, \quad f(y) - f(z) = \langle c, y - z \rangle, \quad e_k := \nabla f(x_k) - c \longrightarrow 0.$$

Let $G(x) = g_0(x) + \langle c, x \rangle$ on S. Convex optimality at z gives $c + \xi + n = 0$ for $\xi \in \partial g_0(z)$, $n \in N_S(z)$, so $G(x) \ge G(z)$ on S. This uses N_S directly, without an unjustified splitting of $N_{K \cap D}$. At every cluster y,

$$G(y) - G(z) = F(y) - F(z). \tag{18}$$

A G-optimal cluster is therefore F-optimal and equals z.

Write $W_k = D_h(z, x_k)$. For any $\epsilon > 0$ there are N and $\delta > 0$ such that

$$k \ge N, \quad W_{k+1} \ge \epsilon \quad \Longrightarrow \quad G(x_{k+1}) - G(z) \ge \delta. \tag{19}$$

Otherwise a convergent subsequence of high-W endpoints with vanishing G-gap would have, by (18), the unique optimal cluster z. Same-point continuity (4) would then force $W \to 0$, a contradiction. This does not assume continuity of $D_h(z, \cdot)$ at other boundary clusters.

With anchor z, (16) becomes

$$W_{k+1} - W_k \le -D_h(x_{k+1}, x_k) - \gamma_{k+1}[G(x_{k+1}) - G(z)] + \gamma_{k+1} \langle e_k, z - x_{k+1} \rangle.$$

Increase N until $\|e_k\| \sup_j \|z - x_j\| \le \delta/2$. Every late endpoint with $W_{k+1} \ge \epsilon$ then satisfies $W_{k+1} < W_k$. An optimal subsequence has $W \to 0$. After a sufficiently late point below ϵ, a first outward crossing to $W \ge \epsilon$ is impossible. Thus $W_k \to 0$; (15) gives $x_k \to z$. ∎

Corollary 7.1. *Theorem 4 applies when h is continuous and strictly convex on all of K, with any closed convex D meeting C. In particular it includes closed affine constraints, $D = \mathbb{R}^n$, and regular composite entropy minimization with*

$$h(x) = \sum_{j=1}^{n} (x_j \log x_j - x_j), \qquad K = \mathbb{R}^n_+.$$

For entropy, f may be any convex C^1 function on an open convex neighborhood of K, subject to local relative smoothness and attainment, and g_0 any finite convex function. Strictness of neither component nor their sum is required.

Proof. Restriction preserves continuity and strict convexity. For entropy,

$$D_h(z, x) = \sum_j [z_j \log(z_j/x_j) - z_j + x_j],$$

with summand x_j for $z_j = 0$. Each summand is nonnegative, tends to zero as $x_j \to z_j$, and tends to infinity as $x_j \to \infty$. This verifies the boundary assumptions. Entropy is 1-coercive and Legendre, strictly convex and continuous on K, with interior Hessian $\operatorname{diag}(1/x_j)$. All hypotheses follow. ∎

This corollary includes the smooth-neighborhood case $g_0 = 0$, every attained orthant-constrained positive-semidefinite quadratic, and the mixed-boundary example $f(u, v) = \frac{1}{2}[(u + v - 1)^2 + u^2]$, whose limit is $(0, 1)$. Its exact updates are

$$u_{k+1} = u_k e^{-\gamma_{k+1}(2u_k + v_k - 1)}, \qquad v_{k+1} = v_k e^{-\gamma_{k+1}(u_k + v_k - 1)}.$$

These earlier special cases are included as genuine corollaries, without inheriting the false general boundary separation claim. A feasible unconstrained stationary point is not needed in this regular-neighborhood corollary. Convex C^2 objectives satisfy local relative smoothness on compact interior sets; C^1 alone is not asserted to imply it.

7.1 A cone that removes the flat feasible face

For the kernel (7), let $D = \{(x, y) : |x| \le y\}$, $f(x, y) = y$, $g_0 = 0$. Then

$$S = \{(x, y) : 0 \le y \le \varepsilon,\ |x| \le y\}$$

meets the bottom edge only at $(0, 0)$. The kernel is continuous and strictly convex on S. Indeed, every nontrivial segment not wholly in the top edge has an open segment in C and hence positive second derivative along it. On the top edge, the second tangential derivative is $\varepsilon b''(x) > 0$. There are no two distinct bottom points in S. Thus Theorem 4 applies although h is not strictly convex on K.

For every $z_0 \in C \cap D$ and positive initial steps, the original *ambient* algorithm converges to $(0, 0)$. It never stops at a finite index because its unique minimizer is on the kernel boundary. No invariant curve, growth-branch choice, or geometric stepsize law from the unconstrained example is asserted for this constrained problem.

8 Bounded-ball strong convexity: a separate sufficient geometry

Continuity of h on its whole closed domain is not silently added to the next result.

Theorem 5. *In the regular setting with $D = \mathbb{R}^n$, replace feasible-set continuity and strictness by*

$$\langle \nabla h(x) - \nabla h(y), x - y \rangle \geq m_R \|x - y\|^2 \quad (x, y \in C \cap B(0, R)), \qquad m_R > 0 \tag{20}$$

for every $R > 0$. Then every original undamped run converges to a minimizer. Equivalently, for C^2 kernels, one may require $\nabla^2 h \succeq m_R I$ on each such bounded interior region, with the bound uniform up to finite boundary points. No global closure continuity is additionally required.

Proof. Lemma 2 supplies boundedness, an optimal cluster, and the retained dissipation. Integrating (20) along interior segments gives $D_h(u, v) \geq (m_R/2)\|u - v\|^2$. This extends to boundary first arguments: along the segment from $v \in C$ to $u \in K$, convexity and lower semicontinuity give convergence of the kernel values to $h(u)$, so passage to the endpoint is valid. The difference-of-merits argument in Theorem 4, now with this quadratic lower bound, proves uniqueness of the optimal cluster without Fact 1.2(iii).

On the bounded orbit ∇f and every ∂g_0 selection are bounded. There are no constraint normals because $D = \mathbb{R}^n$. Pairing (14) with the displacement gives

$$\|x_{k+1} - x_k\| \leq C\gamma_{k+1}.$$

Finite total steps imply a Cauchy orbit. Infinite total steps and (13) give common cluster gradients by the old-endpoint excursion argument, using the displayed displacement bound. The limiting objective G, the new-endpoint inequality, and the high-level barrier (19) then give $D_h(z, x_k) \to 0$ exactly as before. The quadratic lower bound proves norm convergence. Only same-point continuity at z is used in the barrier, so no continuity of $D_h(z, \cdot)$ on the entire orbit closure has been introduced. ∎

Example 1. The nonentropy kernel $h(x) = \sum_j (x_j^2 - \sqrt{x_j})$ on $K = \mathbb{R}^n_+$, extended by infinity outside, satisfies Theorem 5, with $m_R = 2$. For $q(t) = t^2 - \sqrt{t}$,

$$q''(t) = 2 + \frac{1}{4t^{3/2}}, \qquad D_q(a, b) = (a - b)^2 + \frac{(\sqrt{a} - \sqrt{b})^2}{2\sqrt{b}}.$$

This verifies reverse level boundedness and same-point boundary continuity, including $D_q(0, b) = b^2 + \sqrt{b}/2$. The singular derivative at zero gives essential smoothness, and the quadratic growth gives 1-coercivity.

Example 2. The closure-strict result also includes kernels with no bounded-ball strong-convexity modulus. On $(0,\infty)$ define $h(1) = h'(1) = 0$ and

$$h''(x) = w(x) = x^2 + \frac{(1+\sin(1/x))^2}{x}.$$

Set $h(0) = \int_0^1 tw(t)\,dt$ and $h = +\infty$ on $(-\infty, 0)$. The oriented integral $h(x) = \int_1^x (x - t)w(t)\,dt$ tends to $h(0)$ by dominated convergence, since $tw(t) \le t^3 + 4$ near zero. It is strictly convex on the closure: any affine nontrivial segment would contain a positive point where $h'' > 0$. Expanding $(1+\sin s)^2$ gives

$$h'(x) = -\tfrac{3}{2}\log(1/x) + O(1) \to -\infty \quad (x \downarrow 0),$$

using bounded oscillatory integrals of $\sin s/s$ and $\cos(2s)/s$. Thus it is Legendre. The convexity of $h(x) - x^4/12$ gives $h(x) \ge x^4/12 - x/3 + 1/4$, proving 1-coercivity. Finally

$$D_h(0,y) = \int_0^y tw(t)\,dt = \frac{y^4}{4} + \int_0^y (1+\sin(1/t))^2\,dt,$$

which lies between $y^4/4$ and $y^4/4 + 4y$. This verifies both boundary hypotheses. But at $x_j = (3\pi/2 + 2\pi j)^{-1}$, $h''(x_j) = x_j^2 \to 0$, so (20) fails on every neighborhood of zero. With $f(x) = x$, $g_0 = 0$, the boundary minimizer is uniquely zero and Corollary 7.1 applies. This example establishes strict enlargement beyond the bounded-ball curvature condition, without asserting that either general hypothesis automatically implies the other.

9 Entropy with only a continuous boundary gradient

Some positive statements assume only a continuous boundary gradient. This is weaker than a C^1 extension to an open neighborhood, so these statements are recorded separately.

Here $K = \mathbb{R}^n_+$, $C = \mathbb{R}^n_{++}$, and h is entropy. Assume f is proper closed convex, finite and continuous on K, and locally relatively smooth in C. Assume its interior gradient has a finite continuous extension $q : K \to \mathbb{R}^n$. For $u, v \in K$, approximation from the interior gives the supporting inequality

$$f(u) \ge f(v) + \langle q(v), u - v\rangle. \tag{21}$$

This is support on K, not an assertion of a unique ambient subgradient at its boundary.

Theorem 6. *Let $g = 0$ and $S = \operatorname{argmin}_K f \ne \varnothing$. Suppose, for every $z \in S$,*

$$a_z(x) = f(z) - f(x) - \langle q(x), z - x\rangle, \qquad \{x \in K : a_z(x) = 0\} \subset S.$$

Then every original undamped entropy run converges to a minimizer. It suffices that this zero-set inclusion hold only at the cluster points of the given trajectory for its optimal cluster z.

Proof. The energy calculation (12) uses only interior gradients and boundary objective values, so it remains valid. The boundedness and optimal-subsequence argument of Lemma 2 uses only local interior regularity and continuity on K, and also remains valid. Entropy's strict continuous closure geometry and the difference-of-merits argument supply a unique optimal cluster z, without the disputed source fact. The function a_z is continuous nonnegative by (21) and has finite weighted sum.

On a box $[0, M]^n$ containing the orbit, q is bounded. The exact exponential update and the logarithmic mean imply

$$\|x_{k+1} - x_k\| \le C\gamma_{k+1},$$

because for positive u, v, $|u-v| \le \max(u, v)|\log u - \log v|$. For every $\epsilon > 0$, sufficiently late points outside the $\epsilon/2$-ball about z have $a_z \ge \eta > 0$. Otherwise a convergent subsequence outside that ball would give a zero-valued cluster, hence another optimal cluster. A late point outside the ϵ-ball must eventually return to the inner ball because z is a cluster point. Up to its first return, every displacement is charged at an old endpoint with $a_z \ge \eta$, giving

$$\epsilon/2 \le (C/\eta) \sum_{i \ge k} \gamma_{i+1} a_z(x_i) \longrightarrow 0,$$

a contradiction. This proves convergence without needing a stepsize lower bound or controlling a large jump that starts near z. ∎

Strict convexity of $f|_K$ implies the zero-set condition: equality of its support at x and z forces affinity on $[x, z]$. Thus the strict-convex boundary-gradient theorem is a corollary, including positive-definite quadratics with nonstationary boundary minimizers. If f is C^1 on an open convex neighborhood and has a feasible unconstrained stationary minimizer, the condition also holds with multiple minimizers: a support touching such a minimizer must have its zero gradient. This includes feasible stationary positive-semidefinite quadratics and the nonquadratic example $(u+v-1)^4 + w^4$, whose minimizer set is $\{u + v = 1, u, v \ge 0, w = 0\}$. The full regular-neighborhood result of Corollary 7.1 does not need this zero-set condition.

Theorem 7. *Under the continuous-boundary-gradient hypotheses above, let $g : \mathbb{R}^n \to \mathbb{R}$ be finite convex, let $F = f + g$ attain its minimum on K, and assume $F|_K$ is strictly convex. Then every original undamped entropy run converges to its unique minimizer. Neither component needs to be strictly convex separately, and g may be nonsmooth.*

Proof. Coercivity and the sum-rule argument in Lemma 2 establish interior unique proximal points. Their selected subgradients satisfy

$$\log x_{k+1} - \log x_k = -\gamma_{k+1}[\nabla f(x_k) + \xi_{k+1}], \qquad \xi_{k+1} \in \partial g(x_{k+1}).$$

On a box containing the bounded orbit, every ∂g selection is bounded, since a finite convex function is locally Lipschitz. The logarithmic-mean inequality therefore gives

$\|x_{k+1} - x_k\| \le C\gamma_{k+1}$. Boundedness and an optimal subsequence follow by the same valid energy argument.

In the exact source identity used for (12), retain also the nonnegative new-endpoint slack

$$S_g(z, x_{k+1}; \xi_{k+1}) = g(z) - g(x_{k+1}) - \langle \xi_{k+1}, z - x_{k+1} \rangle.$$

Its coefficient is $-\gamma_{k+1}$ and the cross-term estimate does not change it. Hence both

$$\sum_k \gamma_{k+1} a_z(x_k) < \infty, \qquad \sum_k \gamma_{k+1} S_g(z, x_{k+1}; \xi_{k+1}) < \infty$$

hold. Define on the box

$$b_z(x) = \min_{\xi \in \partial g(x)} [g(z) - g(x) - \langle \xi, z - x \rangle].$$

The subgradient graph on the box is compact: it is bounded by local Lipschitzness and closed by the support inequality. Thus the minimum exists and b_z is nonnegative lower semicontinuous. Indeed along a sequence realizing its lower limit, minimizing subgradients have a convergent subsequence, whose limit is an admissible subgradient at the limit point. The actual slack dominates b_z.

If the total steps are finite, the displacement bound gives a Cauchy orbit identified by its optimal subsequence. If the sum is infinite, every cluster y has $a_z(y) = b_z(y) = 0$. Here are both endpoint-sensitive arguments. For a positive old-endpoint cost near y, an arbitrarily late visit to a small inner ball either never leaves a larger ball, contradicting the infinite remaining weight, or makes a first exit with fixed positive displacement. All old endpoints in that excursion are charged, contradicting the summable tail. For a positive new-endpoint cost, choose a late point outside the larger ball and a later visit to the inner ball. Start at the *last* outside index before that visit; all subsequent endpoints are inside and charged, again forcing a fixed positive tail cost. If eventually every point lies inside, infinite weight is already a contradiction. These arguments include a single large exiting or entering jump.

At a cluster y, the equality $a_z(y) = 0$ makes f affine on $[y, z]$ by (21). A minimizing subgradient attaining $b_z(y) = 0$ similarly makes g affine on that segment. Thus F is affine there, and strict convexity forces $y = z$. A bounded sequence with only this cluster converges. ∎

Component strictness (strict f or strict g) is a corollary of Theorem 7, with all boundary-gradient assumptions retained. The larger scope is genuine: $f(x) = (x_1 + 1)^2/2$ and $g(x) = (x_2 - 1)^2/2 + \lambda|x_2 - 1|$, $\lambda > 0$, are individually nonstrict on $\mathbb{R}^2_+$, while their sum is strictly convex and has boundary minimizer $(0, 1)$.

10 Cumulative steps, affine splits, and compatible flat segments

The next results return to the regular ambient setting of Lemma 2, but allow flat feasible boundary segments. Assume h is continuous on K; do not assume closure or feasible-set strictness.

Theorem 8. *Every infinite nonstopped original undamped run has* $\sum_{k\geq 1}\gamma_k = \infty$.

Proof. Suppose the sum is finite and fix $w \in C \cap D$. Boundedness and (16), which does not require strictness, bound $D_h(w, x_k)$ by summation. The interior-anchor dual-ball estimate in Lemma 2 makes $p_k = \nabla h(x_k)$ bounded, hence traps $x_k = \nabla h^*(p_k)$ in a compact subset of C. This conclusion is derived under the contradiction hypothesis; no normal or dual bound was assumed.

On a slightly enlarged compact interior set, Hessians of h have uniform upper and positive lower bounds, ∇f is Lipschitz, and g_0 has bounded subgradients. Comparing the proximal objective with its feasible old iterate gives

$$D_h(x_{k+1}, x_k) \leq B\gamma_{k+1}\|x_{k+1} - x_k\|,$$

so displacements tend to zero because $\gamma_k \to 0$. The uniform Taylor expansions in Lemma 2, now along the entire trapped tail, show $\alpha_k \to 1$, $\Lambda_k \to 1$, and $\gamma_k\ell_k \to 0$. The ratio/extrapolation bound follows from the unchanged first branch. Thus the second branch tends to infinity, and eventually $\gamma_{k+1} = \sqrt{1+\rho_k}\gamma_k > \gamma_k$. This contradicts the finite sum. The proof does not assert that its positive-part bracket is eventually exactly zero. ∎

Theorem 9. *If f is affine on its neighborhood of K, then $F(x_k)$ decreases to m, and every cluster point is a minimizer. In particular a unique composite minimizer implies full primal convergence, with no added strictness of the kernel on its boundary.*

Proof. The primal step is exactly a Bregman proximal-point step for F. Comparing with x_k gives

$$F(x_{k+1}) + D_h(x_{k+1}, x_k)/\gamma_{k+1} \leq F(x_k).$$

For any minimizer z, the three-point inequality gives

$$D_h(z, x_{k+1}) - D_h(z, x_k) \leq -D_h(x_{k+1}, x_k) - \gamma_{k+1}[F(x_{k+1}) - m].$$

Thus weighted endpoint gaps are summable. Theorem 8 and monotonicity force their limit to zero. Closedness and continuity identify every cluster as optimal; uniqueness then gives convergence. This is a proximal-point subclass, not a claim of a new general proximal-point mechanism. ∎

For example, with (7), $f(x,y) = y$ and $g_0(x,y) = \lambda x^2$ for any $\lambda > 0$, the unique minimizer is $(0,0)$ and every initialized undamped run converges there. There is no finite termination because all finite iterates have $y > 0$. This changes the objective of the nonconvergent example and does not invalidate that example. No growth-branch formula or rate is asserted here.

Theorem 10. *Suppose additionally that whenever $u, v \in S$ obey*

$$h((u+v)/2) = [h(u) + h(v)]/2,$$

the restriction of f to $[u,v]$ is affine. If z is any optimal cluster point, then

$$D_f(z, x_k) \to 0, \qquad \nabla f(x_k) \to \nabla f(z).$$

Every cluster y satisfies $\nabla f(y) = \nabla f(z)$ and $f(y) - f(z) = \langle \nabla f(z), y - z \rangle$. This conclusion does not assert that every cluster is optimal, even when the composite minimizer is unique.

Proof. On a compact convex $Q \subset S$ containing the orbit and clusters, put $a(x) = D_f(z,x)$ and use the continuous nonnegative Jensen gap J. If $J(u,v) = 0$, convexity makes h affine on $[u,v]$, and compatibility makes f affine there. Its ambient differentiable supports then give $\nabla f(u) = \nabla f(v)$ and $a(u) = a(v)$. Therefore for any $\eta > 0$, compactness makes

$$j_a(\eta) = \min\{J(u,v) : u, v \in Q,\ |a(u) - a(v)| \geq \eta\} > 0$$

whenever the indicated set is nonempty. Combine $D_h(u,v) \geq 2J(u,v)$ with the feasible-anchor estimate (17). Any change of at least η in a costs at least $2j_a(\eta)/T$ in cumulative steps.

If arbitrarily late $a(x_p) \geq 2\eta$, either $a \geq \eta$ forever thereafter, contradicting Theorem 8 and (13), or a first later index q has $a(x_q) < \eta$. Every old endpoint before that exit has $a \geq \eta$, so this excursion contributes at least $2\eta j_a(\eta)/T$ to a summable tail, again impossible. Hence $a(x_k) \to 0$. At any cluster y, the support of f at y touches at z; open-neighborhood differentiability identifies their gradients. Compactness then promotes the cluster gradient identity to convergence of the full gradient sequence. ∎

The remaining gap in a general unique-minimizer theorem is real at the level of this argument: equal smooth gradients and vanishing reverse divergences do not separate points on flat kernel faces, and the nonsmooth part can still vary there. Neither Theorem 10 nor unique optimality alone closes this implication. The general regular nonlinear split with a flat boundary kernel and unique minimizer remains unresolved by these results.

11 Exact scalar entropy rates

These rates are independent direct recurrence results; they do not rely on the disputed boundary theorem. In this section and the next, write

$$A = \frac{\gamma_1}{\Phi} \prod_{j=2}^{\infty} \frac{\rho_j}{\Phi} > 0, \qquad r = \log \Phi, \tag{22}$$

when the growth recursion is proved. The symbol A here is the stepsize amplitude, distinct from the vertical amplitude in Theorem 1.

Theorem 11. *Let $h(u) = u\log u - u$ on $[0,\infty)$ with its usual extension, $g = 0$, and let f be finite continuous convex on $[0,\infty)$, differentiable on $(0,\infty)$, with $p(u) = f'(u) > 0$ there. The literal undamped recurrence from any $u_0, \gamma_0, \gamma_1 > 0$ is well defined, has strictly decreasing $u_k \to 0$, and satisfies exactly*

$$u_k = u_{k-1}e^{-\gamma_k p(u_{k-1})}, \qquad \rho_{k+1} = \sqrt{1+\rho_k}, \qquad \gamma_k = A\Phi^k[1 + O(\Phi^{-2k})].$$

The curvature candidate is infinite at every adaptive index. The constant A in (22) depends only on the two initial steps.

If $p(u) \sim au^m$ as $u \downarrow 0$, with $a, m > 0$, put $C = [r/(maA\Phi)]^{1/m}$. Then

$$\gamma_k p(u_{k-1}) \to r/m, \qquad u_k \sim C\Phi^{-k/m},$$
$$p(u_k) \sim aC^m\Phi^{-k}, \qquad f(u_k) - f(0) \sim \frac{aC^{m+1}}{m+1}\Phi^{-k(m+1)/m},$$
$$D_h(0,u_k) = u_k.$$

If instead $p(u) \to b > 0$, then

$$\log(1/u_k) \sim \frac{bA\Phi}{\Phi-1}\Phi^k, \qquad \frac{u_{k+1}}{u_k} \to 0, \qquad f(u_k) - f(0) \sim bu_k,$$

while $p(u_k) \to b$ and $D_h(0,u_k) = u_k$. The logarithmic equivalent is not a multiplicative primal equivalent. If additionally $p(u) - b = O(u^\eta)$ for some $\eta > 0$, then for some $C_b > 0$,

$$u_k \sim C_b \exp\left(-\frac{bA\Phi}{\Phi-1}\Phi^k\right).$$

Proof. Convex differentiability makes p continuous nondecreasing. The entropy proximal first-order equation gives the stated unique positive update; zero cannot minimize because the right derivative there is $-\infty$. At a fixed adaptive index set

$$v = u_{k-1}, \quad w = u_k = vq, \quad \beta = \gamma_k p(v) > 0, \quad q = e^{-\beta}, \quad c = p(w)/p(v) \in (0,1].$$

Direct calculation gives

$$D_h(w,v) = v[1-(1+\beta)q], \qquad D_h(v,w) = v[\beta - 1 + q],$$
$$\Delta_h(w,v) = v\beta(1-q), \qquad \gamma_k\ell_k = 1 - c,$$
$$H_k(w) - H_k(v) = -\beta c.$$

Since $h^*(t) = e^t$, the ordered conjugate divergence equals $w[e^{-\delta\beta c} - 1 + \delta\beta c]$. For $t > 0$,

$$0 < e^{-t} - 1 + t = \int_0^t (t-s)e^{-s}\,ds < t^2/2.$$

Consequently, for every $\delta > 0$,

$$0 < \Lambda_k < \frac{q\beta c^2}{1-q} = \frac{c^2\beta}{e^\beta - 1} < c = 1 - \gamma_k \ell_k.$$

Both primal divergences are positive, so the full second branch, including its $\alpha_k/(1+\alpha_k)$ factor, is exactly infinite. This proves the literal growth law. The rationalized ratio identity and positive product proof in Theorem 1 give the stepsize estimate (22); in particular $\gamma_k - A\Phi^k = O(\Phi^{-k})$ is absolutely summable. If u_k decreased to $L > 0$, then $p(u_{k-1}) \geq p(L) > 0$ while $\gamma_k \to \infty$, contradicting that limit. Hence $u_k \to 0$.

For the power law write $p(u) = au^m H(u)$, $H(u) \to 1$, and put $z_k = m\gamma_k p(u_{k-1})$. Exact substitution, without differentiating an asymptotic expression, gives

$$z_{k+1} = a_k z_k e^{-z_k}, \qquad a_k = \rho_{k+1} \frac{H(u_k)}{H(u_{k-1})} \longrightarrow \Phi.$$

Eventually $3/2 \leq a_k \leq 19/10$, so one step later $0 < z_k \leq B = 19/(10e) < 1$. For a sufficiently late N, let $d = \min\{z_N, \log(3/2)\} > 0$. Since te^{-t} increases on $[0, B]$, $[d, B]$ traps the tail. The continuous increasing limit map $F(t) = \Phi t e^{-t}$ then gives

$$\liminf z_k = F(\liminf z_k), \qquad \limsup z_k = F(\limsup z_k).$$

Both limits are positive and thus equal r. It follows that $\gamma_k p(u_k) \to r/(m\Phi)$ and $\gamma_k u_k^m \to r/(ma\Phi)$. These give the primal and gradient equivalents. Integration of p, bounding H between $1 \pm \epsilon$ near zero, gives the objective equivalent without differentiation of H.

For $p(u) \to b > 0$, telescope logarithms:

$$\log(u_0/u_k) = \sum_{j=1}^{k} \gamma_j p(u_{j-1}), \qquad T_k := \sum_{j=1}^{k} \gamma_j \sim \frac{A\Phi}{\Phi - 1}\Phi^k.$$

A weighted-average argument gives the logarithmic equivalent. Also $\gamma_{k+1} p(u_k) \to \infty$, proving the ratio limit, and integration gives $f(u) - f(0) \sim bu$.

Under the extra error bound, $p(u) \geq b$ implies $u_k \leq u_0 e^{-bT_k}$. Thus

$$E_\infty = \sum_{j \geq 1} \gamma_j [p(u_{j-1}) - b] < \infty.$$

Also $R_\infty = \sum_{j \geq 1} (\gamma_j - A\Phi^j)$ converges absolutely, and

$$T_k = \frac{A\Phi}{\Phi - 1}\Phi^k - \frac{A\Phi}{\Phi - 1} + R_\infty + o(1).$$

The exact logarithm identity yields the multiplicative equivalent with

$$C_b = u_0 \exp[bA\Phi/(\Phi - 1) - bR_\infty - E_\infty] > 0.$$

∎

The theorem is an exact recurrence result under its stated scalar hypotheses. To place every member inside all source regularity assumptions, additionally require local relative smoothness, for example local Lipschitz continuity of p on $(0,\infty)$. Convex differentiability alone is not claimed to imply that regularity.

For $f(u)=u^2/2$, set $\beta_k=\gamma_k u_{k-1}$. Then the exact autonomous positive controller is

$$(\beta_{k+1},\rho_{k+1})=\big(\sqrt{1+\rho_k}\,\beta_k e^{-\beta_k},\sqrt{1+\rho_k}\big),$$

and every positive state converges to (r,Φ). Thus it has no nontrivial positive cycles or nonconvergent positive trajectories. With $C=r/(A\Phi)$,

$$u_k\sim C\Phi^{-k},\qquad f(u_k)\sim (C^2/2)\Phi^{-2k},\qquad |f'(u_k)|=D_h(0,u_k)=u_k.$$

The terminal amplitude is independent of u_0 for fixed initial steps, but no uniform transient bound is claimed.

12 Coupled and marginal entropy rates

Consider entropy on $\mathbb{R}^2_+$, $g=0$ and $f(x)=x^TQx/2$. We treat only the two specified positive-entry SPD matrices below, not a matrix-family classification.

Lemma 3. *For either matrix in Theorems 12 and 13, all positive starts and positive initial steps produce finite positive, coordinatewise strictly decreasing iterates. The full curvature candidate is infinite at every step, and*

$$x_{k,i}=x_{k-1,i}e^{-\gamma_k(Qx_{k-1})_i},\quad \rho_{k+1}=\sqrt{1+\rho_k},\quad \gamma_k=A\Phi^k[1+O(\Phi^{-2k})].$$

For $z_k=\gamma_{k+1}x_k$ the exact normalized recurrence is

$$z_{k+1}=a_k z_k\exp(-Qz_k),\qquad a_k=\rho_{k+2}\to\Phi, \tag{23}$$

where exponential and multiplication are coordinatewise.

Proof. At one step write $v=x_{k-1}$, $w=x_k$, $\beta_i=\gamma_k(Qv)_i>0$, $q_i=e^{-\beta_i}$, and $c_i=(Qw)_i/(Qv)_i\in(0,1)$. Set

$$S=\sum_i v_i\beta_i(1-q_i)>0,\qquad W_i=v_i\beta_i(1-q_i)/S.$$

Then $\sum W_i=1$, $1-\gamma_k\ell_k=\sum W_ic_i$, and $H_k(w)-H_k(v)=-(\beta_ic_i)_i$. Since $h^*(p)=\sum e^{p_i}$, the exact ordered conjugate formula and the scalar exponential remainder give

$$0<\Lambda_k<\sum_i W_i\frac{c_i^2\beta_i}{e^{\beta_i}-1}<\sum_i W_ic_i=1-\gamma_k\ell_k.$$

All directed entropy divergences are positive, so the literal branch conclusion follows. The product estimate gives A, and direct substitution gives (23). No autonomous approximation has replaced the actual recurrence. ∎

Theorem 12. *For* $Q = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ *and every positive source initialization,*

$$z_k \longrightarrow (r/3, r/3), \qquad x_{k,i} \sim \frac{r}{3A\Phi^{k+1}}, \qquad \frac{x_{k,1}}{x_{k,2}} \to 1,$$

$$f(x_k) \sim \frac{r^2}{3A^2\Phi^{2(k+1)}}, \qquad D_h(0, x_k) \sim \frac{2r}{3A\Phi^{k+1}}.$$

After a finite start-dependent transient, $\log(z_{k,1}/z_{k,2})$ *preserves its sign and converges geometrically to zero.*

Proof. Eventually $8/5 \le a_k \le 17/10$. Since $te^{-2t} \le 1/(2e)$, after one step both coordinates are at most $B = 17/(20e) < 1/3$. Also $\log(8/5) \ge 3/8 > B$. At a sufficiently late N, put

$$d = \min\{z_{N,1}, z_{N,2}, [\log(8/5) - B]/2\} > 0.$$

The increasing function te^{-2t} on $[0, B]$ and the other-coordinate bound imply $z_{k+1,i} \ge (8/5)de^{-2d-B} \ge d$ whenever both current coordinates lie in $[d, B]$. Thus this rectangle traps the full tail.

Write $u_k = z_{k,1}$, $v_k = z_{k,2}$, $D_k = \log(u_k/v_k)$. Exactly

$$D_{k+1} = D_k - (u_k - v_k) = [1 - L(u_k, v_k)]D_k,$$

where the logarithmic mean L lies in $[d, B] \subset (0, 1)$. Hence $|D_k| \le (1-d)^{k-N}|D_N|$ and eventual sign preservation holds. In particular $u_k - v_k \to 0$. The first-coordinate recurrence is then $u_{k+1} = \Phi u_k e^{-3u_k} + o(1)$. This limit map is continuous increasing on $[d, B]$ because $3B < 1$. Its only positive fixed point is $r/3$, so the liminf/limsup argument gives $u_k \to r/3$ and then $v_k \to r/3$. Divide by γ_{k+1} and use $f(t,t) = 3t^2$, $D_h(0, x) = x_1 + x_2$ to obtain every displayed equivalent. ∎

Theorem 13. *For* $Q = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$, *write* $z_k = (u_k, v_k)$. *For every positive source initialization,*

$$u_k \to r, \qquad kv_k \to 1, \qquad \frac{u_k + v_k - r}{v_k^2} \to -\frac{1}{r}.$$

In particular

$$x_{k,1} \sim \frac{r}{A\Phi^{k+1}}, \qquad x_{k,2} \sim \frac{1}{Ak\Phi^{k+1}},$$

$$k\frac{x_{k,2}}{x_{k,1}} \to 1/r, \qquad k(r - \gamma_{k+1}x_{k,1}) \to 1,$$

$$f(x_k) \sim \frac{r^2}{2A^2\Phi^{2(k+1)}}, \qquad D_h(0, x_k) \sim \frac{r}{A\Phi^{k+1}}.$$

The normalized autonomous limit has eigenvalues $1 - r$ *and* 1 *at* $(r, 0)$.

Proof. The exact nonautonomous equations are

$$u_{k+1} = a_k u_k e^{-u_k - v_k}, \qquad v_{k+1} = a_k v_k e^{-u_k - 2v_k}.$$

Eventually $8/5 \le a_k \le 17/10$, and hence $u_k \le B = 17/(10e) < 1$, $v_k \le B/2$ after one further step. The common multiplier cancels:

$$\log(v_{k+1}/u_{k+1}) - \log(v_k/u_k) = -v_k.$$

Thus $R_k = v_k/u_k$ decreases to $L \ge 0$. If $L > 0$, telescoping implies $\sum v_k < \infty$, hence $v_k \to 0$ and $u_k \to 0$. But then $u_{k+1}/u_k \to \Phi > 1$, incompatible with positive $u_k \to 0$. Therefore $L = 0$, and boundedness gives $v_k \to 0$.

Let $c = \log(8/5)$ and choose a late N with $v_k \le c/2$. Set $d = \min\{u_N, c/2\} > 0$. The map te^{-t} increases on $[0, B]$, so $u_{k+1} \ge (8/5)de^{-d-c/2} \ge d$ whenever $u_k \in [d, B]$. This traps u_k away from zero. The recurrence becomes $u_{k+1} = \Phi u_k e^{-u_k} + o(1)$, whose limit map is increasing on $[d, B]$. Its only positive fixed point is r, giving $u_k \to r$.

Put $\epsilon_k = \log a_k - r = O(\theta^k)$, $\theta = \Phi^{-2}$, and $s_k = u_k + v_k - r$. The exact sum equation is

$$s_{k+1} = (r + s_k)e^{\epsilon_k - s_k} - r + v_k e^{\epsilon_k - s_k}(e^{-v_k} - 1).$$

Define $B(s) = ((r+s)e^{-s} - r)/s$ for $s \ne 0$ and $B(0) = 1 - r$. Then exactly

$$\begin{aligned} s_{k+1} &= B(s_k)s_k - v_k^2 C_k + E_k, \\ C_k &= e^{\epsilon_k - s_k}(1 - e^{-v_k})/v_k \to 1, \\ E_k &= (e^{\epsilon_k} - 1)(r + s_k)e^{-s_k} = O(\theta^k). \end{aligned}$$

Also $v_{k+1}/v_k = e^{\epsilon_k - s_k - v_k} \to 1$. For every fixed $q \in (0,1)$ this gives $v_k \ge c_q q^k$ on a sufficiently late tail. Choose $\sqrt{\theta} < q < 1$; then $\theta^k/v_k^2 \to 0$. Thus the controller's exponentially small perturbation is rigorously negligible at the marginal scale.

Divide the exact sum identity by v_{k+1}^2. For $T_k = s_k/v_k^2$ it gives

$$T_{k+1} = b_k T_k + d_k, \qquad b_k \to 1 - r \in (0,1), \qquad d_k \to -1.$$

The elementary contraction-plus-vanishing-error argument yields $T_k \to -1/r$: after subtracting that fixed value the recurrence has coefficients uniformly below one in modulus and an error tending to zero. Hence $s_k = -v_k^2/r + o(v_k^2)$, without an assumed center-manifold approximation. Finally

$$\frac{1}{v_{k+1}} - \frac{1}{v_k} = \frac{e^{v_k + s_k - \epsilon_k} - 1}{v_k} \longrightarrow 1.$$

Telescoping and Cesaro averaging give $kv_k \to 1$. This also gives $u_k = r - v_k - v_k^2/r + o(v_k^2)$ and $k(r - u_k) \to 1$. Division by γ_{k+1} gives the source-coordinate statements. The normalized objective tends to $r^2/2$, and the entropy distance is the sum of the two coordinates. The limiting Jacobian is $\begin{pmatrix} 1-r & -r \\ 0 & 1 \end{pmatrix}$, proving the eigenvalue statement. ∎

Both fixed quadratics satisfy local relative smoothness: on an interior box with $x_i \le M$, $\nabla^2 h \succeq I/M$, so $L = M\lambda_{\max}(Q)$ suffices.

12.1 Attribution and limitations of the normalized dynamics

The normalized maps are planar competitive Ricker maps. Their autonomous attraction is already studied in [6, 7]; the proof above establishes the actual nonautonomous adaptive trajectory directly rather than claiming a new autonomous attractor. The marginal mechanism and coefficient are also classical. In [8], Section 3.1, equations (7)–(8), the center graph is $h(Y) = -aY - (1-ab)aY^2/K + O(Y^3)$ with reduced map $Y \mapsto Ye^{-Y-bh(Y)}$. Under $X = u$, $Y = 2v$, $K = L = r$, $a = 1/2$, $b = 1$, it gives $u = r - v - v^2/r + O(v^3)$ and $v_{\text{next}} = v - v^2 + O(v^3)$. The reciprocal-increment law $kv \to 1$ is the elementary consequence of that normal form. The present proof verifies the global positive-source-start transfer and its exponentially perturbed adaptive recurrence without importing a formal tracking claim. No novelty or priority claim about the classical mechanism is intended.

13 Conclusions and exact boundaries

The negative and positive results coexist because their geometry differs. The counterexample kernel has a flat optimal boundary face, and reverse divergences to all of its points vanish along a nonconvergent actual trajectory. Feasible-set strictness restores compact separation; bounded-ball curvature supplies a different quantitative route. Regularity of the smooth objective on an open neighborhood is substantive in the common-support argument, while the separate entropy theorems use an explicitly weaker continuous boundary-gradient hypothesis together with their additional zero-set or strictness conditions.

The results establish neither a universal boundary convergence theorem nor a universal unique-minimizer theorem. The undamped exceptional nonconvergent basin in the explicit flat-kernel problem is a measure-zero graph, not an open set of nonconvergent starts. The damped result covers the stated fixed near-one interval, not every damping schedule. The scalar and two fixed-matrix rates do not imply rates for the general positive convergence classes. Finite diagnostic shooting paths for a different oscillatory objective are not used as evidence for any theorem here. Every source-theorem conflict is limited to the inspected arXiv version 2.

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